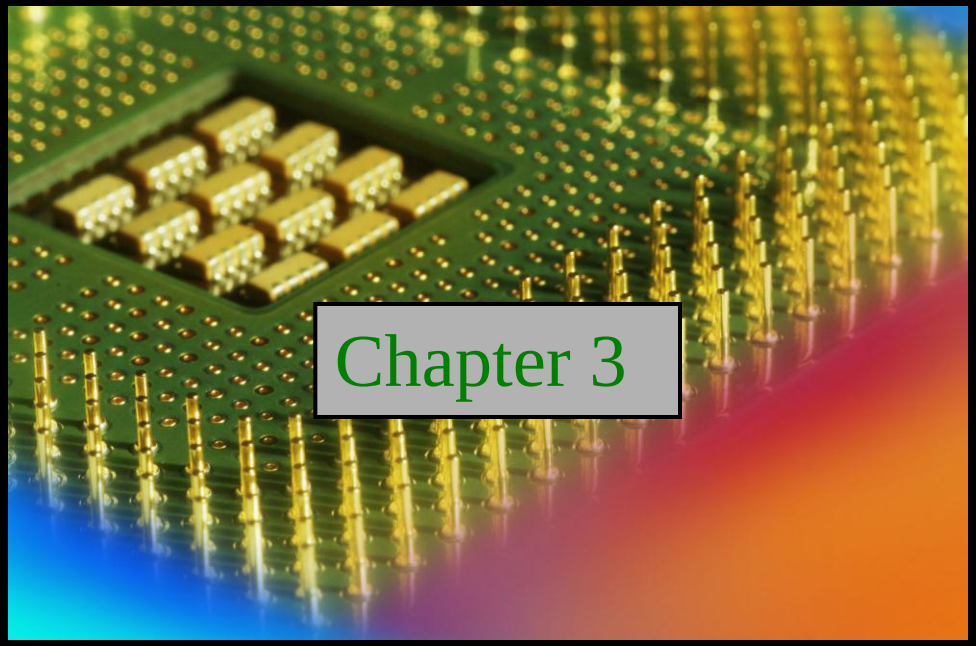
Digital Fundamentals

Tenth Edition

Floyd



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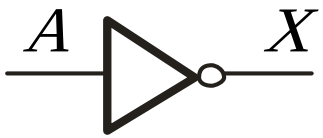
Summary

The Inverter



The inverter performs the Boolean **NOT** operation. When the input is LOW, the output is HIGH; when the input is HIGH, the output is LOW.

Input	Output
A	X
LOW (0)	HIGH (1)
HIGH (1)	LOW(0)

The **NOT** operation (complement) is shown with an overbar. Thus, the Boolean expression for an inverter is $X = \overline{A}$.

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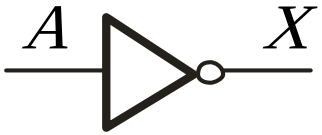
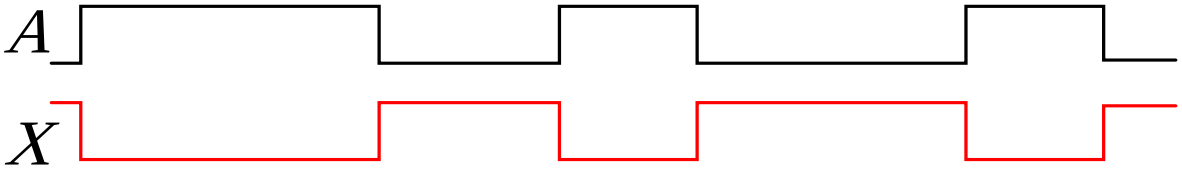
Summary

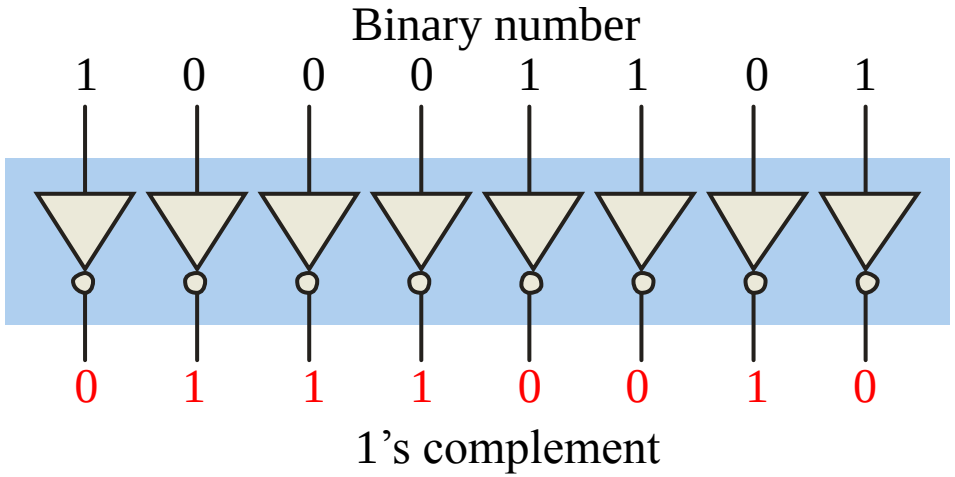
The Inverter



Example waveforms:



A group of inverters can be used to form the 1's complement of a binary number:



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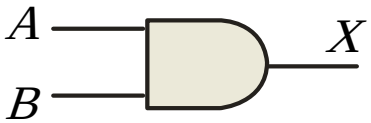
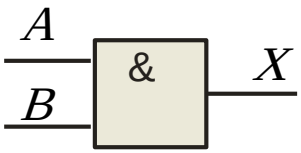
Summary

The AND Gate





The **AND gate** produces a HIGH output when all inputs are HIGH; otherwise, the output is LOW. For a 2-input gate, the truth table is

Inputs		Output
A	*B*	*X*
0	0	0
0	1	0
1	0	0
1	1	1

The **AND** operation is usually shown with a dot between the variables but it may be implied (no dot). Thus, the AND operation is written as $X = A \cdot B$ or $X = AB$.

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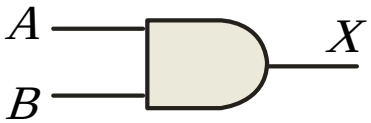
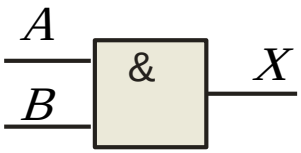
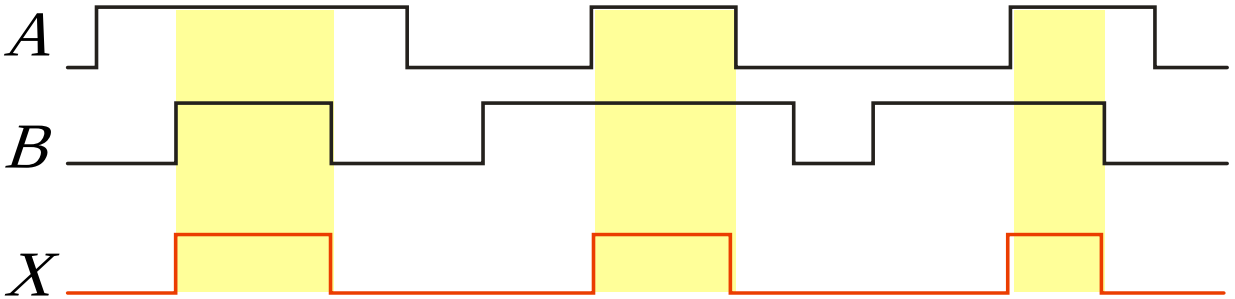
Summary

The AND Gate





Example waveforms:



The AND operation is used in computer programming as a selective mask. If you want to retain certain bits of a binary number but reset the other bits to 0, you could set a mask with 1's in the position of the retained bits.

Example If the binary number 10100011 is ANDed with the mask 00001111, what is the result? 00000011

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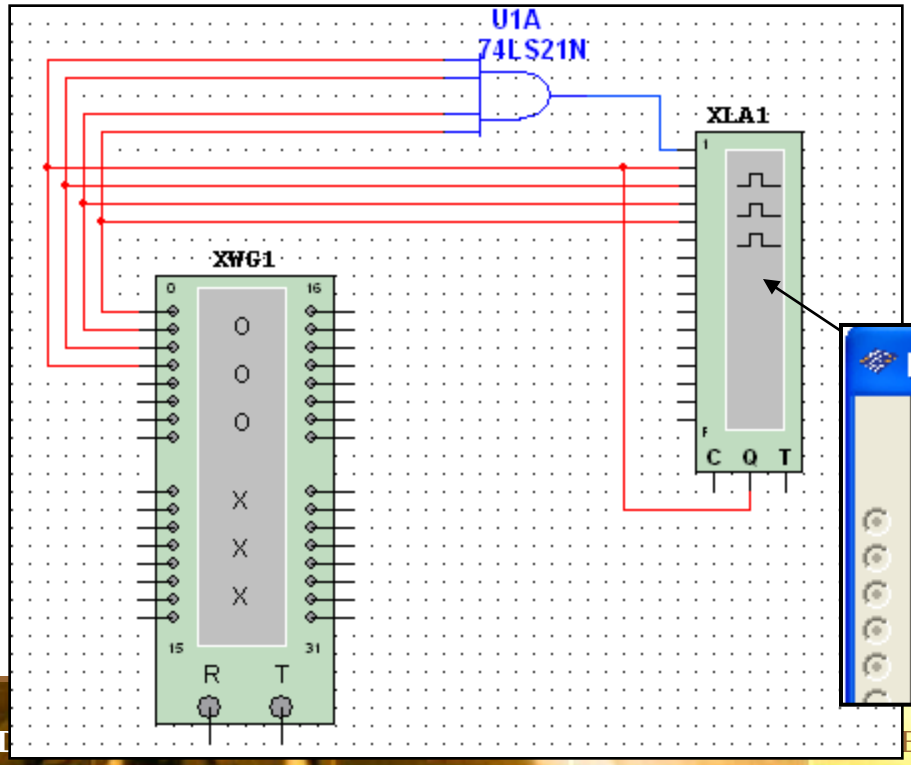
Summary

The AND Gate

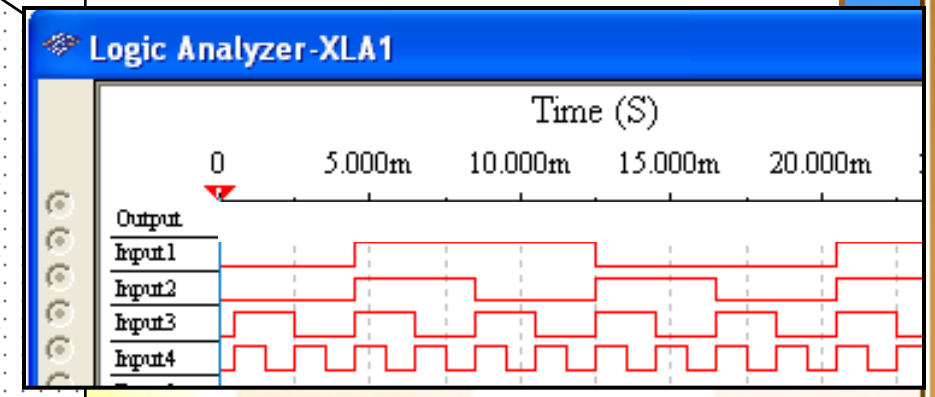
Example

A Multisim circuit is shown. XWG1 is a word generator set in the count down mode. XLA1 is a logic analyzer with the output of the AND gate connected to first (upper) line of the analyzer. What signal do you expect to on this line?



Solution

The output (line 1) will be HIGH only when all of the inputs are HIGH.



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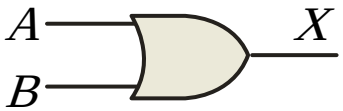
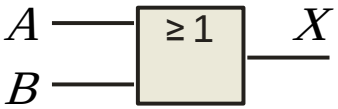
Summary

The OR Gate





The **OR gate** produces a HIGH output if any input is HIGH; if all inputs are LOW, the output is LOW. For a 2-input gate, the truth table is

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	1

The **OR** operation is shown with a plus sign (+) between the variables. Thus, the OR operation is written as $X = A + B$.

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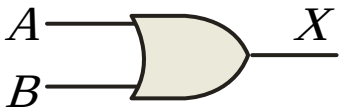
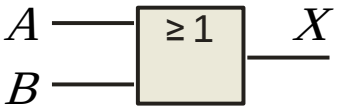
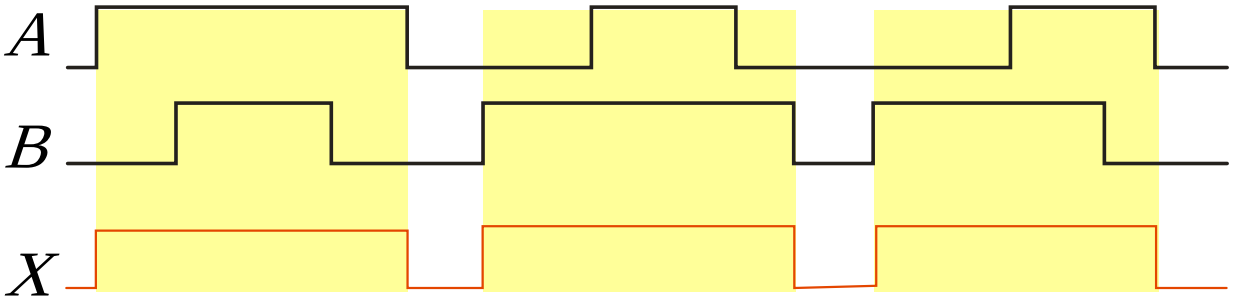
Summary

The OR Gate





Example waveforms:



The OR operation can be used in computer programming to set certain bits of a binary number to 1.

Example ASCII letters have a 1 in the bit 5 position for lower case letters and a 0 in this position for capitals. (Bit positions are numbered from right to left starting with 0.) What will be the result if you OR an ASCII letter with the 8-bit mask 00100000?

Solution The resulting letter will be lower case.

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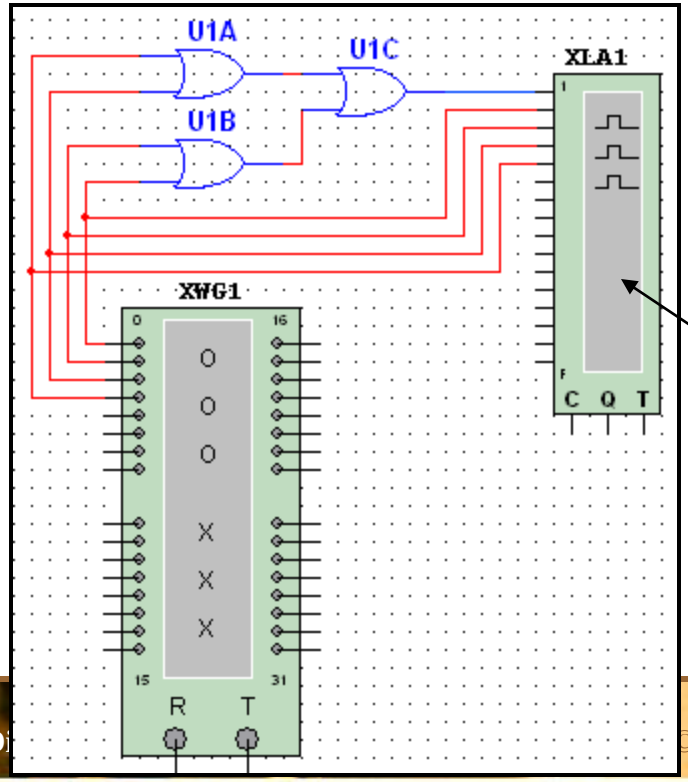
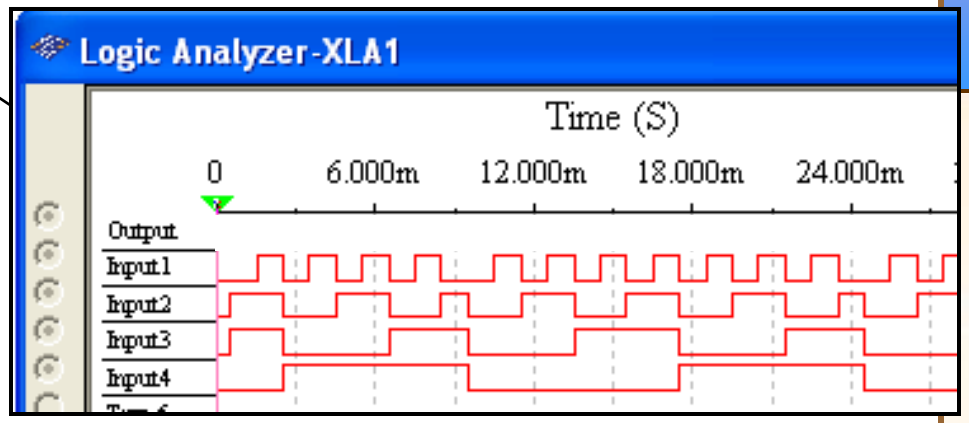
Summary

The OR Gate

Example A Multisim circuit is shown. XWG1 is a word generator set to count down. XLA1 is a logic analyzer with the output connected to first (top) line of the analyzer. The three 2-input OR gates act as a single 4-input gate. What signal do you expect on the output line?



Solution The output (line 1) will be HIGH if any input is HIGH; otherwise it will be LOW.



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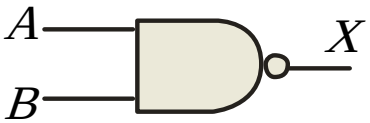
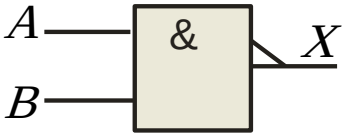
Summary

The NAND Gate





The **NAND gate** produces a LOW output when all inputs are HIGH; otherwise, the output is HIGH. For a 2-input gate, the truth table is

Inputs		Output
A	*B*	*X*
0	0	1
0	1	1
1	0	1
1	1	0

The **NAND** operation is shown with a dot between the variables and an overbar covering them. Thus, the NAND operation is written as $X = \overline{A \cdot B}$ (Alternatively, $X = \overline{AB}$.)

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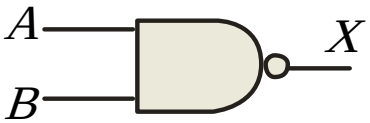
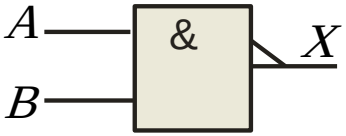
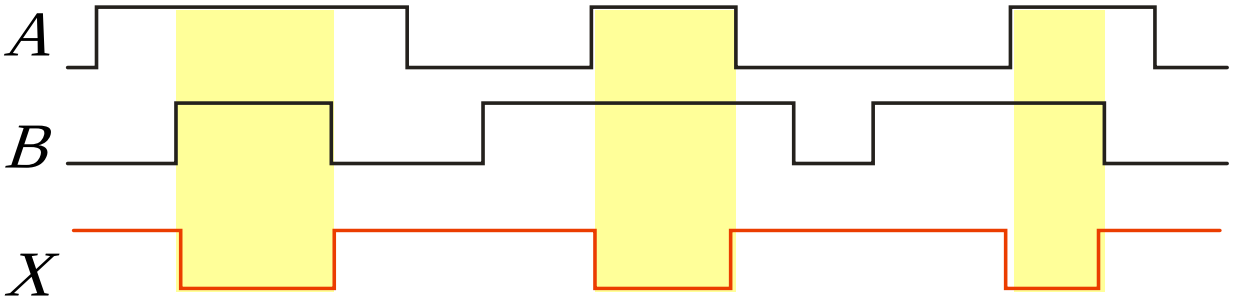
Summary

The NAND Gate



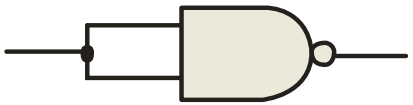


Example waveforms:



The NAND gate is particularly useful because it is a "universal" gate – all other basic gates can be constructed from NAND gates.

Question How would you connect a 2-input NAND gate to form a basic inverter?



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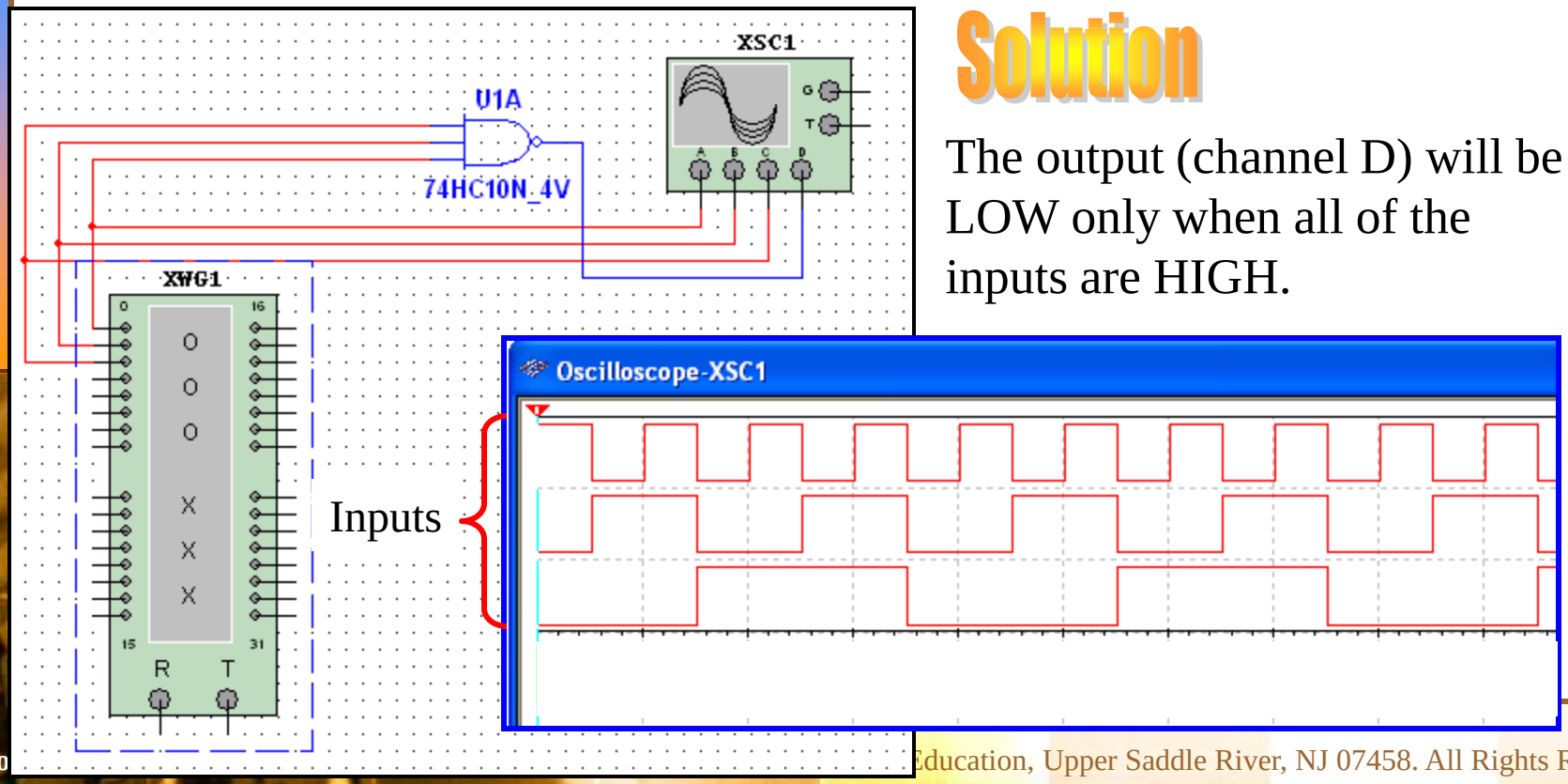
Summary

The NAND Gate

Example

A Multisim circuit is shown. XWG1 is a word generator set in the count up mode. A four-channel oscilloscope monitors the inputs and output. What output signal do you expect to see?



Solution

The output (channel D) will be LOW only when all of the inputs are HIGH.

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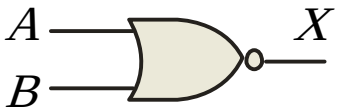
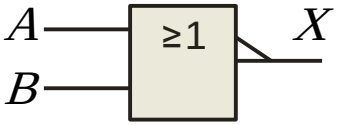
Summary

The NOR Gate





The **NOR gate** produces a LOW output if any input is HIGH; if all inputs are HIGH, the output is LOW. For a 2-input gate, the truth table is

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	0

The **NOR** operation is shown with a plus sign (+) between the variables and an overbar covering them. Thus, the NOR operation is written as $X = \overline{A + B}$.

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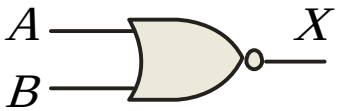
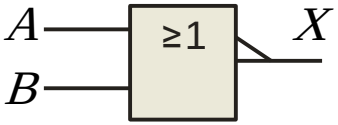
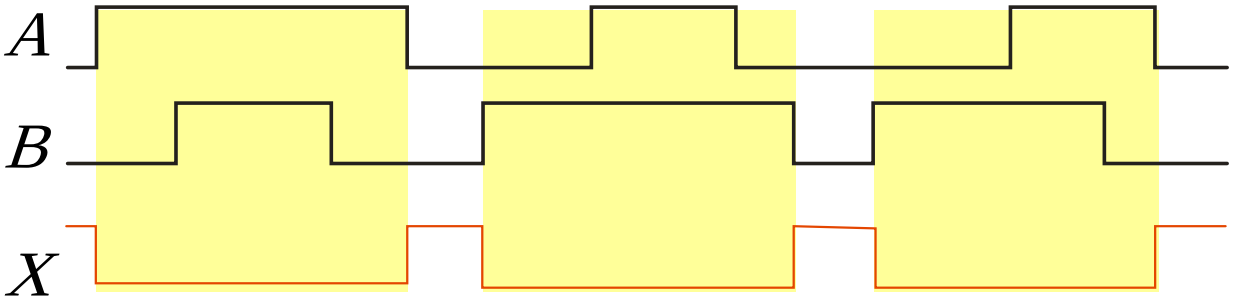
Summary

The NOR Gate



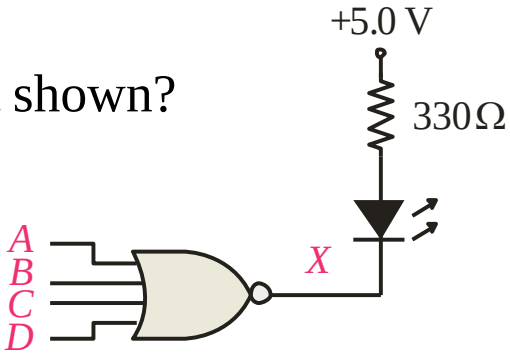


Example waveforms:



The NOR operation will produce a LOW if any input is HIGH.

Example When is the LED is ON for the circuit shown?



Solution The LED will be on when any of the four inputs are HIGH.

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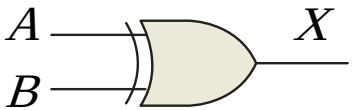
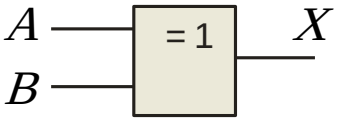
Summary

The XOR Gate





The **XOR gate** produces a HIGH output only when both inputs are at opposite logic levels. The truth table is

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

The **XOR** operation is written as $X = \overline{A}B + A\overline{B}$. Alternatively, it can be written with a circled plus sign between the variables as $X = A \oplus B$.

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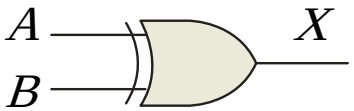
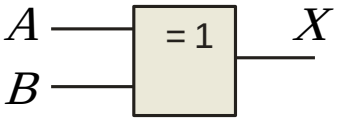
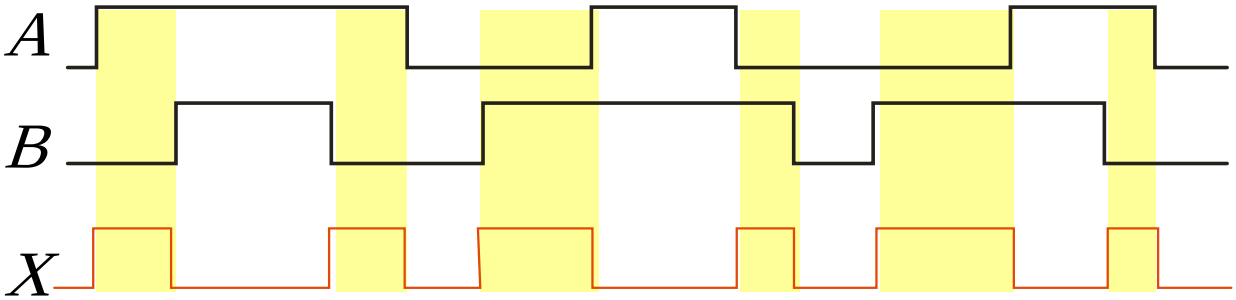
Summary

The XOR Gate





Example waveforms:



Notice that the XOR gate will produce a HIGH only when exactly one input is HIGH.

Question If the *A* and *B* waveforms are both inverted for the above waveforms, how is the output affected?

There is no change in the output.

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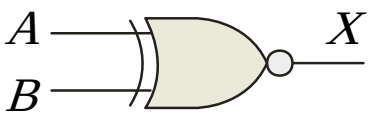
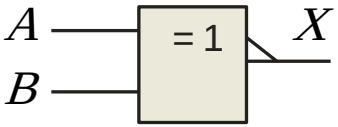
Summary

The XNOR Gate





The **XNOR gate** produces a HIGH output only when both inputs are at the same logic level. The truth table is

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

The **XNOR** operation shown as $X = \overline{A}\,\overline{B} + AB$. Alternatively, the XNOR operation can be shown with a circled dot between the variables. Thus, it can be shown as $X = A \odot B$.

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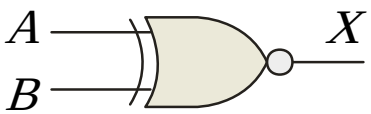
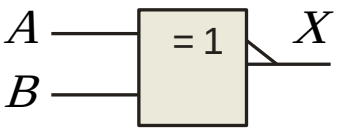
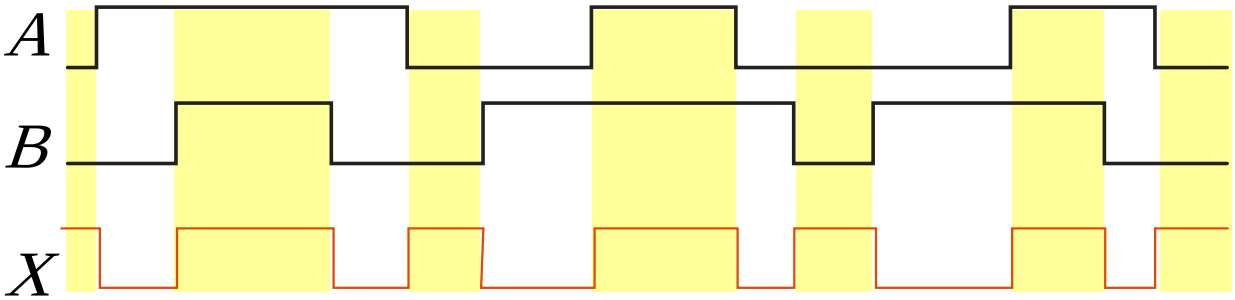
Summary

The XNOR Gate





Example waveforms:



Notice that the XNOR gate will produce a HIGH when both inputs are the same. This makes it useful for comparison functions.

Question If the *A* waveform is inverted but *B* remains the same, how is the output affected?

The output will be inverted.

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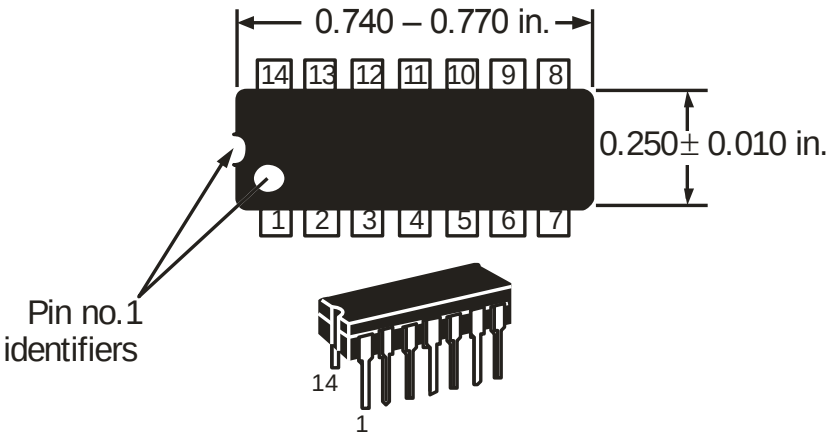
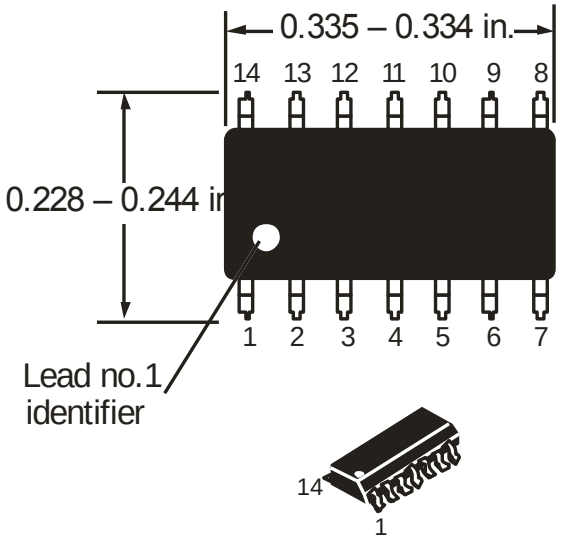
Summary

Fixed Function Logic

Two major fixed function logic families are TTL and CMOS. A third technology is BiCMOS, which combines the first two. Packaging for fixed function logic is shown.



DIP package



SOIC package

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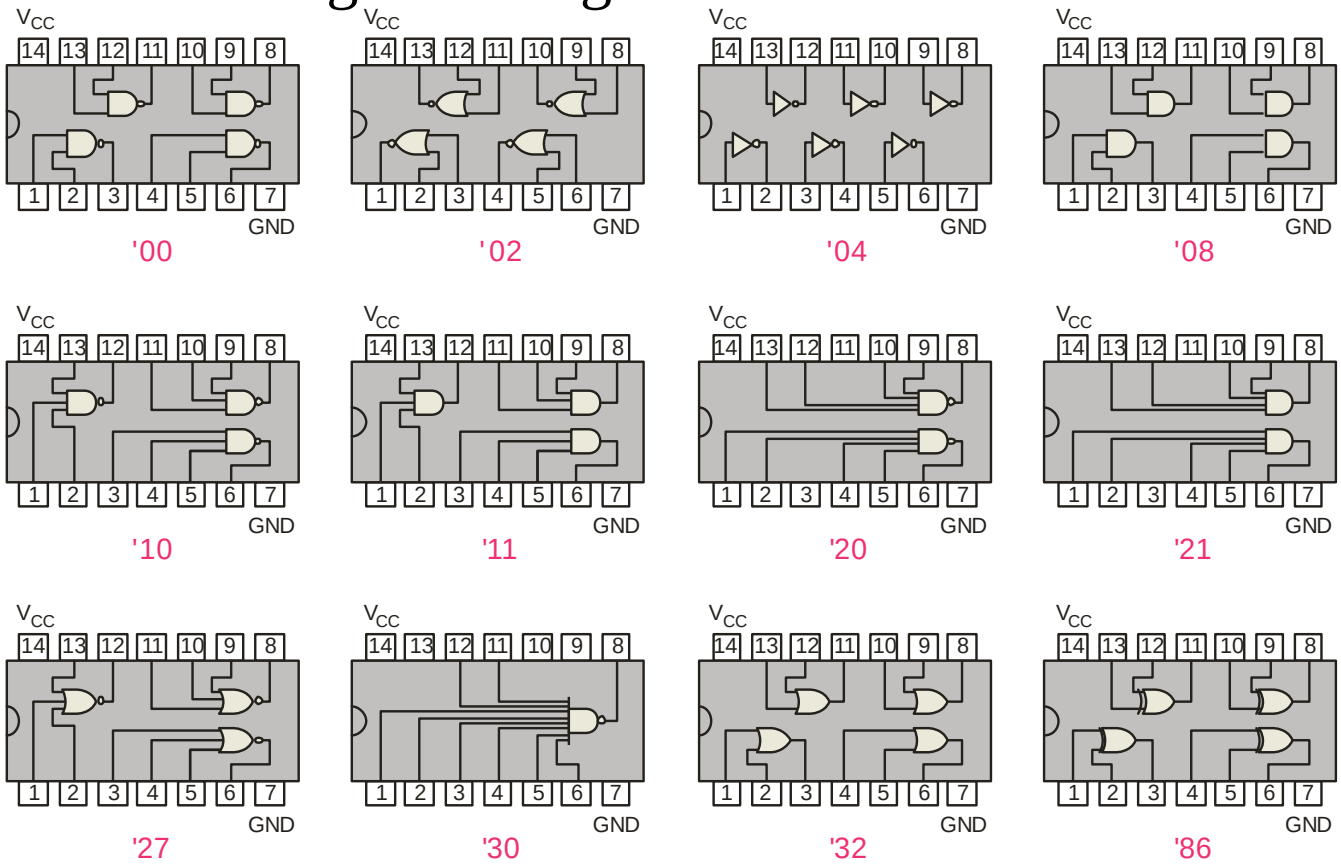
Summary

Fixed Function Logic

Some common gate configurations are shown.



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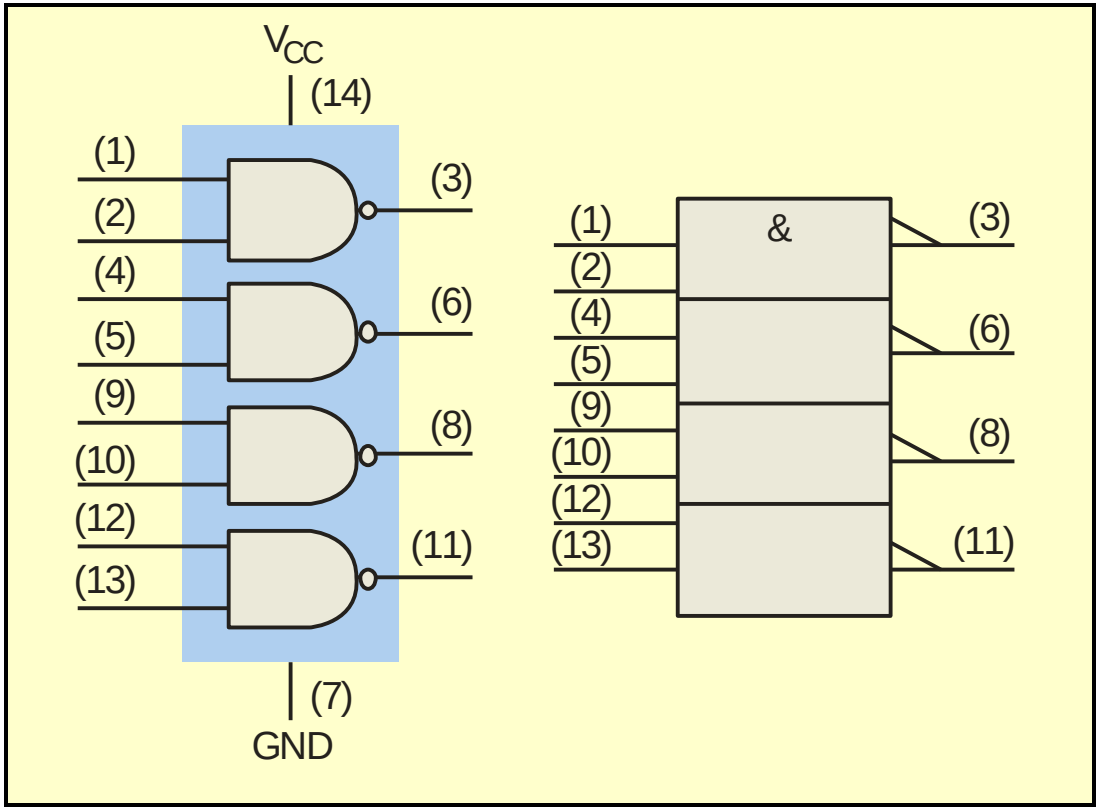
Summary

Fixed Function Logic

Logic symbols show the gates and associated pin numbers.



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Summary

Fixed Function Logic

Data sheets include limits and conditions set by the manufacturer as well as DC and AC characteristics. For example, some maximum ratings for a 74HC00A are:

MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V_{CC}	DC Supply Voltage (Referenced to GND)	–0.5 to + 7.0 V	V
V_{in}	DC InputVoltage (Referenced to GND)	–0.5 to V_{CC} +0.5 V	V
V_{out}	DC Output Voltage (Referenced to GND)	– 0.5 to V_{CC} +0.5 V	V
I_{in}	DC Input Current, per pin	±20	mA
I_{out}	DC Output Current, per pin	±25	mA
I_{CC}	DC Supply Current, V_{CC} and GND pins	±50	mA
P_D	Power Dissipation in Still Air, Plastic or Ceramic DIP † SOIC Package † TSSOP Package †	750 500 450	mW
T_{stg}	Storage Temperature	–65 to + 150	°C
T_L	Lead Temperature, 1 mm from Case for 10 Seconds Plastic DIP, SOIC, or TSSOP Package Ceramic DIP	 260 300	°C

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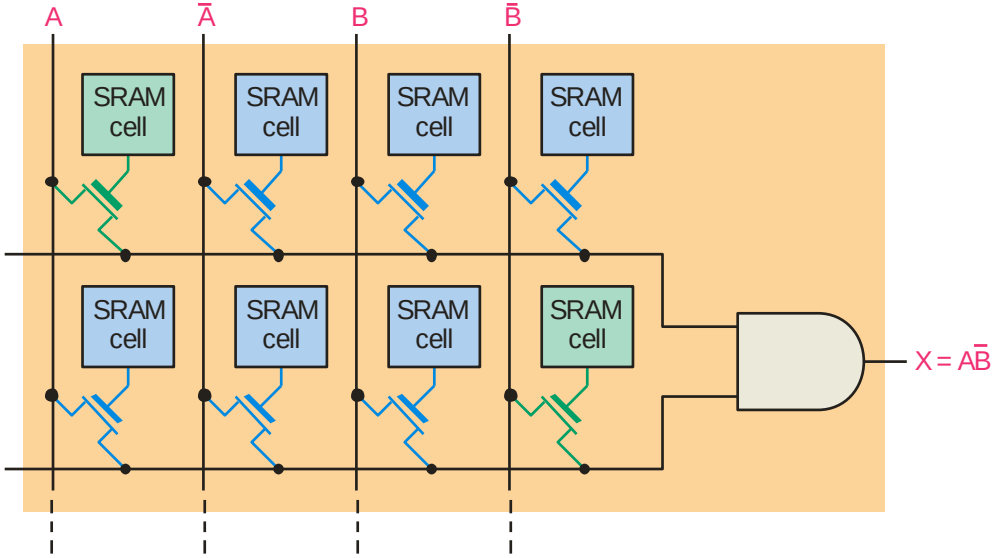
Summary

Programmable Logic

A Programmable Logic Device (PLD) can be programmed to implement logic. There are various technologies available for PLDs. Many use an internal array of AND gates to form logic terms. Many PLDs can be programmed multiple times.



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Summary

Programmable Logic

In general, the required logic for a PLD is developed with the aid of a computer. The logic can be entered using a Hardware Description Language (HDL) such as VHDL. Logic can be specified to the HDL as a text file, a schematic diagram, or a state diagram.

Example

A text entry for a programming a PLD in VHDL as a 2-input NAND gate is shown for reference in the following slide. In this case, the inputs and outputs are first specified. Then the signals are described. Although you are probably not familiar with VHDL, you can see that the program is simple to read.

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Summary

Programmable Logic

```
entity NandGate is
        port(A, B: in bit;
        LED: out bit);
end entity NandGate;
```

```
architecture GateBehavior of NandGate is
signal A, B: bit;
begin
        X <= A nand B;
        LED <= X;
end architecture GateBehavior;
```


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Selected Key Terms

Inverter	A logic circuit that inverts or complements its inputs.
Truth table	A table showing the inputs and corresponding output(s) of a logic circuit.
Timing diagram	A diagram of waveforms showing the proper time relationship of all of the waveforms.
Boolean algebra	The mathematics of logic circuits.
AND gate	A logic gate that produces a HIGH output only when all of its inputs are HIGH.

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Selected Key Terms

OR gate A logic gate that produces a HIGH output when one or more inputs are HIGH.

NAND gate A logic gate that produces a LOW output only when all of its inputs are HIGH.

NOR gate A logic gate that produces a LOW output when one or more inputs are HIGH.

Exclusive-OR gate A logic gate that produces a HIGH output only when its two inputs are at opposite levels.

Exclusive-NOR gate A logic gate that produces a LOW output only when its two inputs are at opposite levels.

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Quiz

1. The truth table for a 2-input AND gate is

a.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	0
1	0	0
1	1	1

d.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	1

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Quiz

2. The truth table for a 2-input NOR gate is

a.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	0
1	0	0
1	1	1

d.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	1

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Quiz

3. The truth table for a 2-input XOR gate is

a.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	*B*	*X*
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	0
1	0	0
1	1	1

d.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	1
1	0	1
1	1	1

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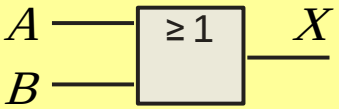
Quiz

4. The symbol 

is for a(n)

a. OR gate

b. AND gate

c. NOR gate

d. XOR gate

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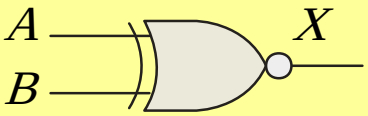
Quiz

5. The symbol 

is for a(n)

a. OR gate

b. AND gate

c. NOR gate

d. XOR gate

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Quiz

6. A logic gate that produces a HIGH output only when all of its inputs are HIGH is a(n)

a. OR gate

b. AND gate

c. NOR gate

d. NAND gate

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Quiz

7. The expression $X = A \oplus B$ means

a. A OR B

b. A AND B

c. A XOR B

d. A XNOR B

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Quiz

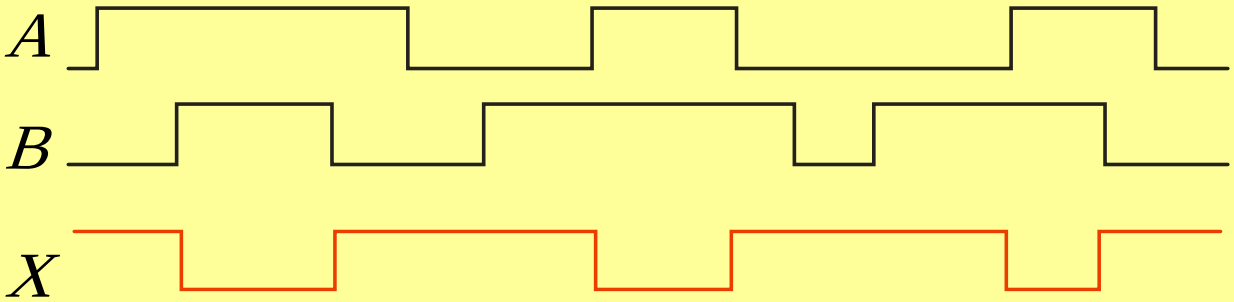
8. A 2-input gate produces the output shown. (*X* represents the output.) This is a(n)

a. OR gate

b. AND gate

c. NOR gate

d. NAND gate



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Quiz

9. A 2-input gate produces a HIGH output only when the inputs agree. This type of gate is a(n)

a. OR gate

b. AND gate

c. NOR gate

d. XNOR gate

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Quiz

10. The required logic for a PLD can be specified in an Hardware Description Language by

a. text entry

b. schematic entry

c. state diagrams

d. all of the above

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Quiz

Answers:

1. c	6. b
2. b	7. c
3. a	8. d
4. a	9. d
5. d	10. d

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